

Here are the detailed, correctly formatted notes for both videos.

Video 1: Expected Values, Main Ideas!!! (Discrete Variables)

Video Link: https://youtu.be/KLs_7b7SKi4

Channel: StatQuest with Josh Starmer

I. Introduction: The Bet

- The video introduces **Expected Value** using a bet scenario in "Statland" with Statsquatch.
- The Bet:** Statsquatch bets you \$1 that the next person you meet *has* heard of the movie "Troll 2".
- This means:
 - If the person **has heard** of "Troll 2", you **lose \$1**.
 - If the person **has NOT heard** of "Troll 2", you **win \$1**.

II. Gathering Data & Probabilities

- A survey of Statland's population (213 people total) reveals:
 - 37 people** *have* heard of "Troll 2".
 - 176 people** *have never* heard of "Troll 2".
- This allows us to calculate the probabilities for each outcome:
 - Probability (Heard of "Troll 2")** = $37 / 213 = 0.17$.
 - Probability (Never heard of "Troll 2")** = $176 / 213 = 0.83$.

III. What is Expected Value?

- Expected Value is the **long-run average** of an event if it were repeated many, many times.
- It helps you decide if the bet is good or bad on average.
- Intuitive Calculation (Assuming 100 Bets):**
 - Expected Losses:** You would expect to lose about 17 times ($100 \text{ bets} \times 0.17$).
 - Total Loss = $17 \times (-1) = -17$.
 - Expected Wins:** You would expect to win about 83 times ($100 \text{ bets} \times 0.83$).
 - Total Win = $83 \times (+1) = +83$.
 - Net Result (after 100 bets):** $+83 - 17 = **+\$66**$.
- Expected Value (Average Gain Per Bet):**
 - Total Net Gain / Number of Bets = $66 / 100 = **0.66**$.
 - This is the **Expected Value** of the bet. It means *on average*, you expect to **gain 66 cents** every time you take this bet.

IV. The General Formula (for Discrete Variables)

- A more direct way to calculate the expected value ($E(X)$) is to multiply the *value* of each outcome by its *probability* and sum the results.
- Formula:**

$$E(X) = (\text{textValue}_1 \times \text{textProb}_1) + (\text{textValue}_2 \times \text{textProb}_2) + \dots$$

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- Applying to the Bet:**
 - Outcome 1 (Heard of "Troll 2"): Value = $-\$1$, Prob = 0.17
 - Outcome 2 (Never heard): Value = $+\$1$, Prob = 0.83
- $\$$ ParseError: KaTeX parse error: Can't use function '\$' in math mode at position 1: $\$E(X) = (-\$1 \times 0.17) + (\$1 \times 0.83) = \0.66

ParseError: KaTeX parse error: Can't use function '\$' in math mode at position 1: $E(X) = -0.17 + 0.83 = +0.66$ ParseError: KaTeX parse error: Can't use function '\$' in math mode at position 1: E

- **Formal Notation:**

$$E(X) = \sum_i x_i \cdot P(x_i)$$

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- $P(x_i)$ is the probability of that outcome x_i .

V. Second Example: A Different Bet

- Statsquatch changes the payout:
 - If they **have heard** of "Troll 2", you **win \$10**.
 - If they **have NOT heard**, you **lose \$1**.

- **Calculation:**

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- **Conclusion:** This is an even better bet for you, with an average expected gain of 87 cents per bet.

Video 2: Expected Values for Continuous Variables!!!

Video Link: <https://youtu.be/OSPr6G6Ka-U>

Channel: StatQuest with Josh Starmer

I. Review: Discrete vs. Continuous

- **Discrete Variables:** Have distinct, separate outcomes (e.g., win/lose, 1, 2, 3, 4). We use **summation** ($\sum$) to find the expected value.
- **Continuous Variables:** Can take on *any* value within a range (e.g., time, height, weight). We must use **integration** ($\int$) to find the expected value.

II. The New Scenario: Continuous Data

- **The Question:** What is the **expected value for waiting time** to meet the next person in Statland?
- "Waiting time" is a continuous variable.
- **Modeling:** Instead of collecting infinite data, we can model the waiting times with a **Probability Density Function (PDF)**. For waiting times, this is often the **Exponential Distribution**.
- **Exponential Distribution Formula:**

$$f(x) = \lambda e^{-\lambda x}$$

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- λ (lambda) is the "rate" parameter (e.g., people met per second). In the video, $\lambda = 0.05$ fits the data.

- **Key Point:** For a continuous variable, the probability of any *single exact value* is zero. We can only find the probability of a *range* by finding the **area under the curve (AUC)** using an integral. The y-axis is "Probability Density" or "Likelihood," not probability.

III. Deriving the Continuous Formula (The Intuition)

1. **Approximate:** We can *approximate* the continuous curve by slicing it into discrete rectangular bars (like a histogram).
2. **Discrete Formula:** We can use the discrete expected value formula, $E(X) = \sum x_i \cdot P(x_i)$, on these bars.
 - x_i = The value of the outcome (e.g., the midpoint of the bar, like 5s, 15s, 25s).
 - $P(x_i)$ = The probability of that outcome (the *area* of the rectangle, which is $\text{Height} \times \text{Width}$).
3. **Refine:** To get a better approximation, we make the rectangles infinitely thin (i.e., the width of the rectangles goes to zero).
4. **The Calculus Connection:** A sum ($\sum$) of infinitely thin rectangles is an **integral** ($\int$).
 - The *width* of the rectangle (Δx) becomes dx .

- The height of the rectangle is the value of the function, $f(x)$.
- The "probability" (area of the rectangle) becomes $f(x)dx$.
- The outcome x_i just stays x .

IV. The General Formula (for Continuous Variables)

- The formula for the expected value of a continuous variable is:

$$E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

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V. Solving for the Exponential Distribution

- We plug the exponential PDF, $f(x) = \lambda e^{-\lambda x}$, into the general formula. Since time can't be negative, our integral is from 0 to ∞ .

$$E(X) = \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx$$

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- This integral is solved using **Integration by Parts**, which results in:

$$E(X) = \frac{1}{\lambda}$$

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- **The Answer:** The expected value for an exponential distribution is simply **1 divided by the rate (λ)**.
- **Applying to the Scenario:**
 - The given rate was $\lambda = 0.05$ (people per second).
 - $E(X) = \frac{1}{0.05} = 20$
 - The expected waiting time between people is **20 seconds**.

http://googleusercontent.com/youtube_content/9 http://googleusercontent.com/youtube_content/10